

Student Performance Analysis

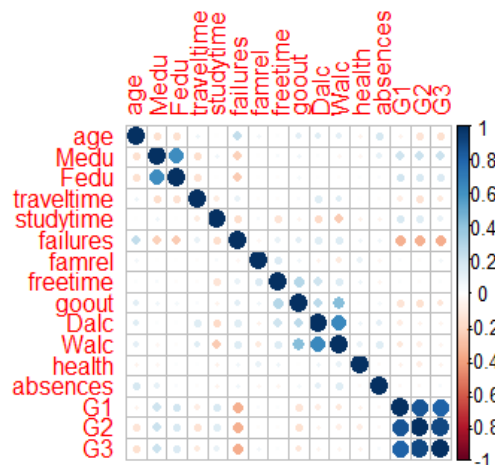
The objective to perform this project is to analyze the student performance using multivariate techniques, like PCA (Principal Component Analysis), Cluster analysis and Factor Analysis. Using these techniques together to find the underlying factors influencing the performance so that we can make the conclusion about the dataset.

Methodology:

- 1) Data Preprocessing : The dataset was cleaned to handle the missing values and numeric value was scaled to ensure that there is uniform impact across the dataset and to perform the multivariate analysis.
- 2) Dimensionality Reduction:
 - PCA was used to capture the variance, with 1st dimension explaining 21.2% of variance.
 - Factor Analysis was performed to reduce complex variables, and find the underlying factors (latent factors) which are affecting the performance. The three latent factors captured are Academic Performance, Lifestyle/Social Behavior, and Socio-Economic Background.
- 3) Clustering:
 - K-means clustering was applied on the dataset to divide the student population into 4 groups, which was found optimal using the elbow method.
 - Hierarchical Clustering (Ward's method) was used to validate the groupings through a dendrogram structure.

Key findings:

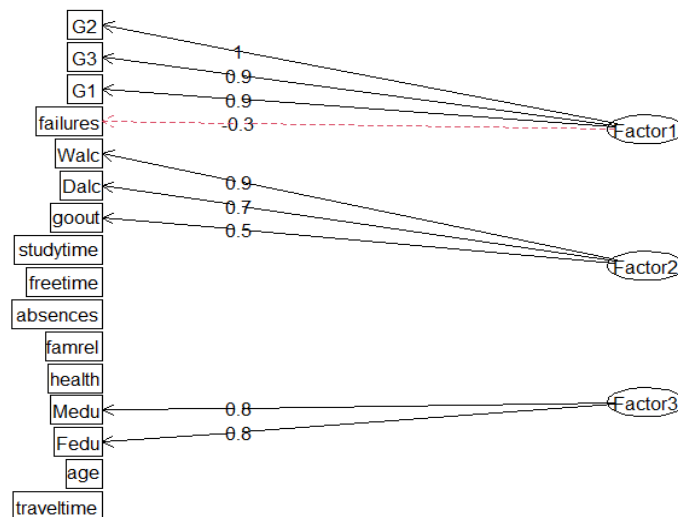
- Academic Predictability: Strong positive correlation between grades G1, G2, and G3 indicates that early academic performance is a robust predictor of final success.



➤ Factor Interpretation :

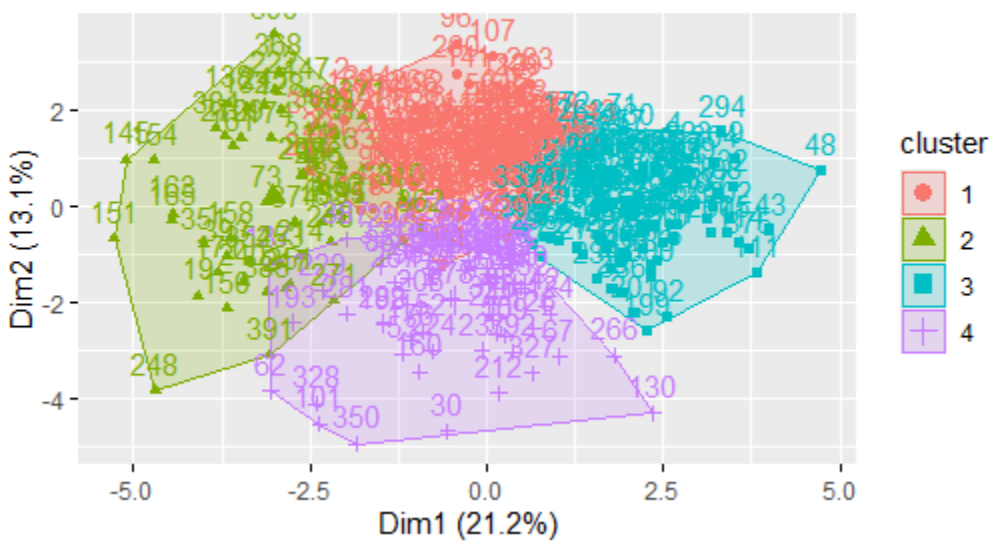
- 1) Factor 1 (Academic) :strongly associated with the grades G1,G2 and G3 and negatively with the failure.
- 2) Factor 2 (Lifestyle) : Depends on the socialization(go out) and alcohol consumption(Dalc,walc)
- 3) Factor 3 (Background) : Depends on parental education (medu,fedu)

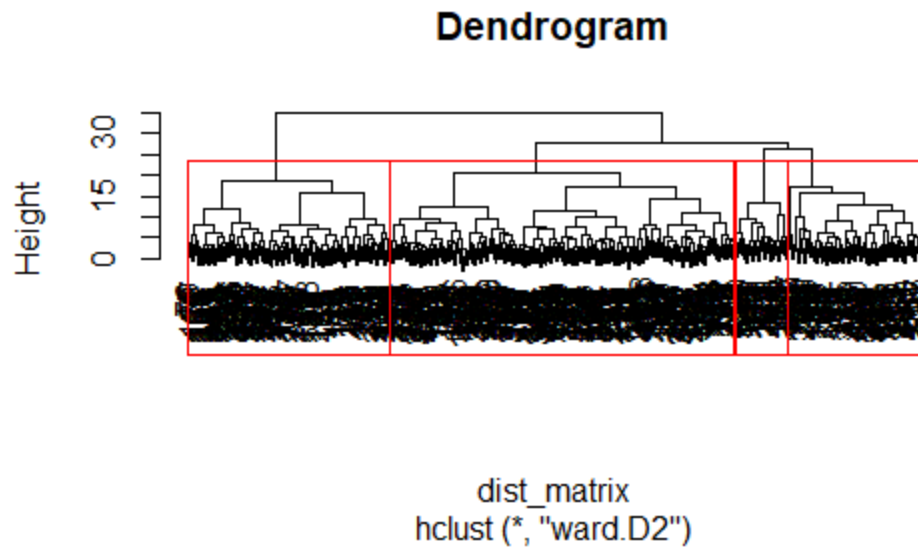
Factor Analysis



- Segmentation :The students were successfully divided into four groups (cluster) ,allowing for personalized supported strategies .

Cluster Visualization





Conclusion:

The analysis confirms that the student performance is not solely depending on the grades but also the lifestyles choices ,social life and background.By identifying these clusters, educational institutions can move beyond one-sized-fits-all strategies and provide targeted support to students who are at risk due to lifestyle factors versus those needing socio-economic assistance.

Recommendations:

- Early Intervention: Students showing a drop in performance between G1 and G2 should be targeted immediately.
- Behavioral Support: Clusters exhibiting high scores in Factor 2 (Lifestyle) could benefit from counseling or wellness workshops.
- Resource Allocation: Students with lower scores in Factor 3 (Background) may benefit from scholarship or mentorship programs.